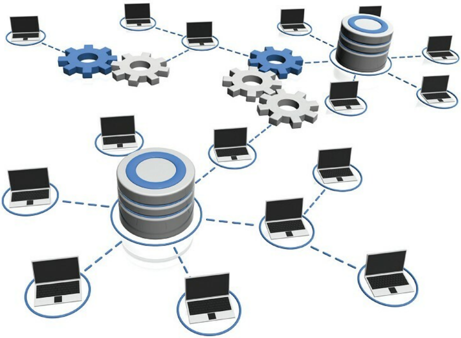
The Best Open Source Databases for IoT Applications

The Internet of Things (IoT), because of its inherent nature, requires certain features in the databases associated with it. This article gives a tiny selection of open source database management systems that are suited to usage for the IoT.



The term 'Internet of Things' is used to refer to: (i) the global network of smart objects interconnected by means of Internet technologies, (ii) the set of supporting technologies necessary to realise this, i.e., RFIDs, sensors, inter-machine communicating devices, and (iii) the ensemble of applications and services leveraging such technologies to open new business and marketing opportunities.

According to a report by Gartner, 8.4 billion interconnected devices will be in use in the world in 2017. The Internet of Things presents highly novel challenges, especially to database management systems, like integrating tons of voluminous data in real-time, processing events as they stream, and dealing with the security of data. An example would be IoT based environment temperature sensors fitted in smart cities, which produce huge amounts of data on the temperature and humidity of the live atmosphere in just a few minutes.

In order to handle IoT data effectively, it is highly important to find the right sort of database. But choosing an efficient database for IoT applications could be really challenging as the IoT environment is not always the same. There are many factors which have to be kept in mind while choosing a database for IoT applications. The most important of these are scalability, ability to handle huge amounts of data at adequate speeds, flexible schema, portability with varied analytical tools, security and costs.

An IoT database should have the capability of being fault-tolerant and highly available. If any node in the database cluster goes down, it should still be capable of accepting read and write requests. Distributed databases make multiple copies or replicas of data, and write the data over multiple servers. If any server storing the data fails, then other servers take over the task of storing and respond to the query till the failed server is up. IoT databases should be highly available, as the IoT database handling systems can face highly voluminous writes and stores. If any database server is down or the data write is too high for a distributed database in real-time, data can be stored in the messaging system until the database processes the backlog of data or any additional servers which are added to the main database cluster.

The following are some of the top open source databases available for IoT based applications.

InfluxDB

InfluxDB is an open source distributed time series database developed by InfluxData. It is written in the Go programming language, and is based on LevelDB, a key-value database. In addition to a front-end, an HTTP interface and libraries are provided to users for database interaction. The main